

JY-CRPG-BENCH: A LONG-HORIZON BENCHMARK FOR FRONTIER MODELS IN AN OPEN-WORLD RPG

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ABSTRACT

A long-horizon task requires a model to perceive its environment, decide for itself what is worth doing, and hold one plan across hours of dependent decisions. We present JY-CRPG-BENCH, which places a frontier model inside Heroes of Jin Yong, a 1996 open-world role-playing game run under DOS emulation. Its ending, recovering the fourteen books of the Jin Yong novels, takes a first-time human player tens of hours. The game requires reading 320×200 isometric pixel art and 16-pixel Traditional Chinese, knowledge of the novels its world is built from, and a keyboard as the only actuator. It has no quest log, marker or score, so the model must maintain its own map and goals, and it must win turn-based fights against the game AI. Progress is scored by milestones of the game itself and, once a model finishes, by completion time. Under a 60-minute budget, in which a human speedrun finishes the game, one of 11 frontier models plays past the opening of the game, and none gains experience or holds a book. Four-hour sessions of 10 models bring one more model to the compass, and still none gains experience.

1 INTRODUCTION

Frontier models solve competition mathematics and write working compilers, yet the tasks they complete reliably are still measured in hours of human work (Kwa et al., 2025). A long horizon is a property of the task, the number of dependent steps its completion requires (Chen et al., 2026), and the benchmarks that measure it have so far run in text, as business simulations (Backlund & Petersson, 2025; Fan et al., 2026) or interactive fiction (Phan et al., 2025). A game adds the requirement they leave out: the model has to recover its own state from raw perception, because nothing in the environment reports it. Existing game benchmarks each remove one part of that problem, by rendering the state as text, removing language, or handing the model its objectives (Paglieri et al., 2025; ARC Prize Foundation, 2026; Ju et al., 2026), so the full combination is untested.

We present JY-CRPG-BENCH, built on Heroes of Jin Yong (金庸群俠傳), a 1996 Chinese role-playing game that we run as its original DOS binary under emulation. Its ending requires recovering the fourteen books of the Jin Yong novels from an open world, a task that takes a first-time human player tens of hours, and Figure 1 shows its world. We chose the game because it tests five capabilities of a frontier model at once.

1. The model must read the screen. The frame is 320×200 pixel art in a 45° isometric projection, and every objective, prompt and statistic on it is Traditional Chinese in 16-pixel glyphs, a script that multimodal evaluation has largely neglected (Tam et al., 2025). No text is supplied to the model.
2. The model must bring knowledge the game never states. The world is assembled from the fourteen novels, so who a character is, which faction to join and which storyline leads to a book are known to a reader of the novels and written nowhere on screen.
3. The model must set its own goals. The game keeps no quest log, draws no marker and reports no score, so the model maintains its own map and plan and decides for itself what is worth doing. ARC-AGI-3 places the same goal-setting requirement at the centre of agentic intelligence (ARC Prize Foundation, 2026).
4. The model must act through a keyboard alone, exposed through an API that it calls as a tool. Movement runs along four diagonals, no key moves straight up the screen, and every system of the game is a menu.



Figure 1: The environment as the model receives it, upscaled three times without interpolation.

5. The model must win fights. Fights are turn-based on a grid against the AI of the game, with position, range and inner force to manage, and a party at level one can lose them.

Together these form one long-horizon task, a persistent world with sparse and delayed reward over thousands of dependent steps. This is the setting of reinforcement learning, and here a frontier model plays it in context, with no training on the game. Two further properties make the game a usable benchmark: it is menu-driven, so a model that deliberates between actions is not penalised as in a real-time game (cf. Zhang et al., 2025a), and its progression state is measurable per character, so progress is scored against events of the game itself.

Table 1 places JY-CRPG-BENCH among interactive agent benchmarks, which the next section reviews. Section 3 details the environment, the interface, the session protocol and the measurement, and Section 4 reports which milestones the models reach and how they play.

Table 1: Interactive agent benchmarks. Observation is what the model receives before any upscaling, a self-directed environment supplies no task list, score or objective marker during play, and the last column is the record that verifies progress.

Benchmark	Environment	Observation	Language	Self-directed	Progress from
JY-CRPG-BENCH	one 1996 CRPG	320×200 px	Trad. Chinese	yes	memory, save, replay
BALROG	6 research games	text, render	English	no	env. reward
TextQuests	interactive fiction	text	English	yes	game state
ARC-AGI-3	abstract puzzles	64×64 grid	none	yes	env. score
VideoGameBench	1990s action games	native px	English	no	checkpoint images
GameWorld	34 browser games	browser	English	no	game state
MineExplorer	Minecraft	3D render	English	no	milestone rules

2 RELATED WORK

2.1 GAME ENVIRONMENTS AS AGENT BENCHMARKS

Games have been the standard setting for sequential decision-making since the Arcade Learning Environment (Bellemare et al., 2013), and language and vision-language agents have produced a second generation of game benchmarks. BALROG aggregates six reinforcement-learning environments under one protocol and reports a persistent gap between what a model knows about a game and what it does inside one (Paglieri et al., 2025). Imgame-Bench isolates how much of a game result belongs to the harness (Hu et al., 2026), and Orak spans twelve popular titles with configurable agentic modules (Park et al., 2026). VideoGameBench is the nearest benchmark to ours, running real 1990s games from raw pixels under a no-scaffold rule (Zhang et al., 2025a). GameWorld standardises the action interface across 34 browser games and pairs every task with a state-verifiable metric (Ouyang et al., 2026). MineExplorer evaluates open-world exploration in Minecraft and finds that models handle single-hop tasks but degrade once hidden prerequisites must be coordinated over a longer trajectory (Ju et al., 2026).

The nearest precedents in horizon are the public Pokémon runs. Claude 3.7 Sonnet reached three gym badges from screen pixels with a memory scaffold, where an earlier model had not left the first house (Anthropic, 2025). Gemini 2.5 Pro finished Pokémon Blue in 813 hours with a harness that translated the RAM of the game into text, and an ablation without vision performed about as well (Comanici et al., 2025). The PokeAgent Challenge turned the setting into a speedrunning track scored by completion time over milestones (Karten et al., 2026a). PokeGym builds a long-horizon benchmark on the open-world game Pokémon Legends: Z-A with no access to game state (Zhang et al., 2026a), and StarBench one on the turn-based Honkai: Star Rail (Zhang et al., 2025b). Crafter established the single-environment, achievement-based design we follow (Hafner, 2022), and NetHack remains the deepest environment in use for learned agents, with runs of tens of thousands of turns (Küttler et al., 2020). Cradle and Voyager drive games through a foundation model with a skill library or a code API (Tan et al., 2024; Wang et al., 2024a).

2.2 LONG-HORIZON EVALUATION AND ITS MEASUREMENT

Vending-Bench established long-term coherence as the property that separates long-horizon tasks from long chains of short ones (Backlund & Petersson, 2025), and E-Commerce Bench extends the setting to a year of deterministic business operation (Fan et al., 2026). Both keep perception out of the loop. TextQuests shares our self-directed, long-horizon framing while removing vision (Phan et al., 2025), and TextAtari stretches horizons to 10^5 steps over text renderings (Li et al., 2025). ARC-AGI-3 isolates adaptation to novel tasks by excluding language and external knowledge (ARC Prize Foundation, 2026), and AutumnBench probes world-model quality through environment-level queries (Warrier et al., 2025). UltraHorizon makes exploration under hidden rules the task itself (Luo et al., 2025), and Kapoor et al. (2026) observe that benchmark design has favoured tasks that are precisely specified, automatically graded and short, and call for open-world evaluation. Navigation is a documented weakness even without perception: one-shot maze navigation fails by ten cells a side for every model tested (Jiang et al., 2026), and from coordinates alone the failures are attributed to looping (Einarsson, 2025).

Long horizons also complicate measurement. The ABC checklist catalogues task-validity and outcome-validity failures across agentic benchmarks (Zhu et al., 2025). An audit of agent traces finds reward hacking in a majority of them on two suites (Shao et al., 2026), and a survey of the horizon gap finds that the field answers uninformative outcome signals by building denser step-level ones (Chen et al., 2026). State-based verification has precedent outside games, where τ -bench compares the final database state with an annotated goal (Yao et al., 2024).

3 JY-CRPG-BENCH

3.1 ENVIRONMENT

Heroes of Jin Yong, published by Heluo Studio (河洛工作室) in 1996, is an open world of two tiers: many local scenes such as buildings, sects and caves attached to one world map of 480×480 tiles

(scarsty & contributors, 2026). Each of 320 named characters carries a record of level, experience, health, stamina, skills, items, reputation and aptitude, and every scene carries an entry condition. Of the 84 scenes, 7 are open at the start: the home, five inns (客棧) and the house of the hermit (南賢). The conversation with the hermit opens all but 4 of the rest, and from there the order of play is left to the player. The only instruction the game gives comes from its fiction: enter the world of the Jin Yong novels, join its factions, learn its martial arts, and recover the fourteen books that end the game.

The game fixes the order of its opening and leaves the campaign open. It starts in a small compound with a searchable chest, one speaking character and one exit tile, and beyond the exit the path south across the world map leads to the hermit. A cabinet beside him holds the compass (羅盤), the only source of coordinates the game offers, and his conversation opens the house of the first companion, who joins after a yes-or-no prompt at which the confirm key means no. The opening ends there and needs no fight. The campaign that follows runs through fights, the experience and levels that victories bring, and the chains of items, factions and fights that lead to the books. Fights are turn-based on the scene grid: each turn a character moves along the four axes and then strikes, uses an item, waits or rests, and skills draw on inner force (內力). The first scripted fight begins with the party drained of it, and a party that goes elsewhere first meets storylines whose fights it cannot survive at level one. Some of these fights end the game when lost, leaving only the load menu, and others let the story continue. With all fourteen books and enough reputation, the ending is a boss fight at a shrine on the far side of the world map (bocelli, 2023).

The two phases test different capabilities. The opening is played by walking, reading and answering, and we give the model the route as far as the compass, so a stall there is a failure of perception or navigation. The campaign adds combat and the knowledge of the novels, since the game names its characters, factions and storylines and explains none of them. Appendix A states the licences of the benchmark code, the emulator and the game.

3.2 OBSERVATION AND ACTION SPACE

The model observes the raw framebuffer at 320×200 and acts by pressing keys. A scored session serves the native frame without upscaling, annotation, set-of-marks overlay or accessibility tree (cf. Xie et al., 2024). The frame combines several documented weaknesses of vision-language models. Low resolution reduces their accuracy (Niu et al., 2025), and images far from the training distribution degrade recognition even of familiar categories (Lin et al., 2026). Text rendered as pixels is understood less well than the same text supplied as tokens (Liu et al., 2026), and precise spatial reading fails once primitives are small or adjacent (Rahmanzadehgervi et al., 2024; Wang et al., 2024b). The action space is the DOS keyboard, with 130 key names over 100 distinct keys, and Figure 5 shows the few the game responds to. Movement runs along the four numpad diagonals, with the arrow keys aliased onto the same axes, so a model that presses up expecting to walk up the screen moves up and to the right.

The single action presses a key, or a list of up to 100 keys in order, holding each for a requested number of emulated frames, and returns once the screen has settled, including after a scene change, so the frame the model receives is the one to act on. There is no call that advances the game without input. A scored session answers four of the calls in Table 2 and refuses the three that save, list and restore emulator snapshots, so a run cannot be rewound from outside the game. The save and load menu of the game stays available to the model, as it is to a human player, so a model that loses a fight can reload the last save the game wrote. Appendix B states the settle rule in frames, and Appendix C gives the reasons for the list action and for the other choices of the protocol.

3.3 HARNESS AND SESSION PROTOCOL

The benchmark supplies the brief and the control surface, and the model plays through an agent harness. Every model in this paper ran inside pi¹, a minimal coding agent, installed without modification and set to its high thinking level. Its four tools read, write and edit files and run a shell, from which the model drives the control surface with HTTP calls. The brief is the whole instruction the model receives. It covers the session call, the calls that constitute play, the opening route as far as

¹<https://github.com/badlogic/pi-mono>

the compass, and a manual of the controls and systems of the game. It is published as `agents.md`, the project file `pi` reads at launch, in English and in Chinese, and both versions keep the on-screen Traditional Chinese terms verbatim. An action returns only its metadata unless the model asks for the frame in the same reply, so the model looks at the screen when it decides to. The benchmark adds no memory scaffold, no map, no pathfinder and no access to the state of the game. A model keeps any memory between actions in files it writes in that shell, and a scored session receives no field that reports its progress until the run ends, so a model cannot query its own score. The catalogue is served only to the leaderboard page, the run store refuses listing, and a container profile runs the harness without network access beyond its own session. The leaderboard therefore ranks models under one fixed harness, since harness choice changes game results (Hu et al., 2026; Karten et al., 2026b) and on long-horizon tasks the variance a harness induces can exceed the variance between models (Zhang et al., 2026b).

A session is created with a declared model name and a playtime of 240 minutes by default and at most twenty-four hours, and its clock starts at the first playable moment. The server keeps the wall clock of every decision call, so played time and think time are measured for any client. Each session is a separate emulator process with a private copy of the game, supplied by the host, with its save slots reset to a new game, so no run inherits the progress of another. A session ends when its budget lapses or the game is completed. When the budget lapses the session renders its recording to a time-compressed video, computes its metrics, and appends itself to the public catalogue with the video and its keypress timeline. Figure 2 shows the apparatus, and Appendix B lists the emulation and hosting parameters.

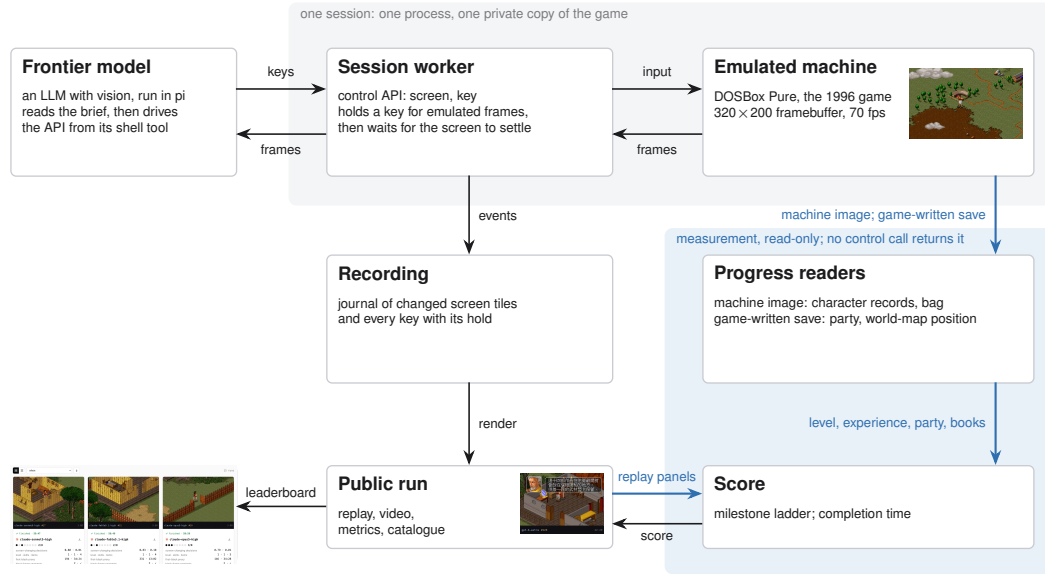


Figure 2: A frontier model in pi drives the control API of its own session. The recording feeds the public replay, and progress is read from the emulated machine, the save the game writes and the replay, none of which the control API exposes.

3.4 MEASURING PROGRESS

Progress is read from three records that the control surface never exposes: the memory of the emulated machine, the save the game writes, and the published replay. The game keeps its persistent state in one archive, with the world position, the party roster and the shared bag in front of the 320 character records, and the bag is the working copy that changes the moment an item is taken. From the live machine the benchmark reads the item count, the compass and the books after every action, and the level and experience of every character. The party roster and the world position are read from the save the game writes. The game offers its save menu on the world map alone and draws it with six rows there and four in a scene, so every two minutes the benchmark presses one escape while the model is idle and saves when the menu has six rows. A successful save is therefore itself

the evidence that the party has crossed onto the world map, and the first one times the crossing. The session is held while the attempt runs, so the model never sees the screen mid-save, and its few seconds of menu time are not charged to the run.

Six events leave no trace in either record, because the game keeps them only on screen. The model enters a scene from the world map, speaks with the hermit, opens the item screen with the compass in it, enters a fight, loses it, or answers the prompt of a companion who asks to join. Each draws fixed art at a fixed screen position, so these events are read from the published replay by matching every frame against that art, and Figure 3 names the panel behind each milestone. The readings do not alter the game: the benchmark serialises the machine and never loads state into it, and the only keys it presses on its own belong to the save attempt. A milestone, once observed, stays credited, so an item picked up and later used still counts. Appendix D gives the archive layout, the save attempt and the panel matching in full and checks every reading against the others wherever two records exist.

The milestone scale has 11 steps, following the achievements of Crafter (Hafner, 2022): six in the opening and five in the campaign. The opening milestones are the exit tile, the chest in the first room, a scene entered from the world map, the conversation with the hermit, the compass in his cabinet, and the companion who joins through dialogue. The campaign milestones are a fight entered, a fight fought to its end, experience gained, the second level, and the first of the fourteen books. The milestones are listed in the order of the opening route, with the optional chest after the exit, and a run is credited with every milestone it reaches, so the count does not depend on the path taken.

The milestones extend to the ending of the game. The fourteen books² are required before the final sequence can be played, and the archive records every book held, so the count of books continues the scale beyond the last milestone. The first reading that finds all fourteen books records the completion time, which ranks the sessions that finish.

4 EVALUATION

We evaluate 11 models from 5 vendors at the 60-minute budget, with 40 sessions on record and 2 to 6 per model. A model is credited with every milestone any of its sessions reached, and a session created under a variant of a model name, a spelling or the thinking level, is listed under the model. The baseline is a seeded random policy that plays the same budget through the same control surface (Mnih et al., 2015; Hafner, 2022; ARC Prize Foundation, 2026). It draws from 13 keys the brief teaches, weighted towards movement and confirmation, and never reads the screen. The hour budget is sized to the opening, and the results are reported as shares of sessions and as spreads within a model.

4.1 MAIN RESULTS

Of the 11 models, all 11 reach the world map, 9 pick up an item, 4 speak with the hermit, and one, gpt-6-astra, takes the compass, recruits a companion, enters a fight and fights it to the end. It reaches 8 of the 11 milestones across its 4 sessions. No model gains experience: every one of the 35 character records read still shows level 1, 0 experience and 1 skill, the values at the start of the game, so every milestone from experience onward remains unreached. Figure 3 shows, for each model, the share of sessions that reached each milestone. Its two top rows are human references read from published videos of the game: 4 speedruns, all of which finish the game in 60 to 126 minutes, and 3 playthroughs by experienced players, one of which reaches the ending. The speedruns cross onto the world map within the first minute in 32 to 107 keypresses, speak with the hermit by minute 2, fight by minute 3 and hold the first book between minutes 26 and 72, and 2 of the 4 take the compass. The playthroughs hold the first book between minutes 58 and 105.

²Item records 144 to 157 of the item table, named after the novels: 飛狐外傳、雪山飛狐、連城訣、天龍八部、射鵰英雄傳、白馬嘯西風、鹿鼎記、笑傲江湖、書劍恩仇錄、神鵰俠侶、俠客行、倚天屠龍記、碧血劍、鴛鴦刀. A book counts as held in the shared bag or carried by the player.

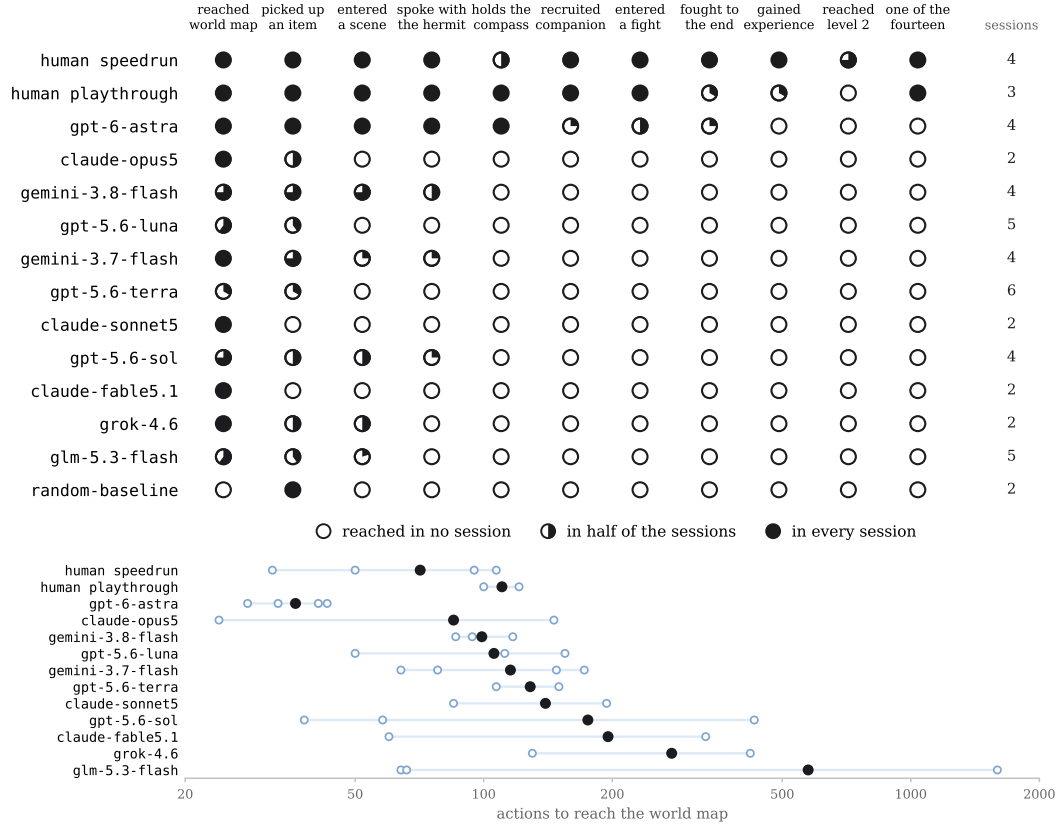


Figure 3: Milestones per model, with rows ordered by the mean number of actions to the world map, fewest first, and the random baseline last. In the upper panel each marker shows the share of sessions in which the model reached the milestone, with the number of sessions at the right. The lower panel shows the actions to the world map across the sessions that crossed, on a log scale, with a light dot per session and a black dot for their mean. The two human rows are read from published videos, with a keypress as the action, and Appendix D gives the record behind each milestone.

4.2 QUALITATIVE ANALYSIS

Four milestones separate the models: the chest, the first scene beyond the home, the hermit and the fight. Figure 4 shows one frame from each behaviour pattern behind these divisions. The random policy reaches 1 of the 11 milestones in 60 minutes, since a random walk in a small room reaches the chest within the hour. The key sequence recomputed from its seed matches the recorded timeline key for key, and no save succeeds during its hour, so the world map is the first milestone a random walk does not reach. The chest therefore separates models from one another but not from the baseline: 9 models searched it and 2 never did.

Every model crosses onto the world map, in 30 of the 40 sessions, and 6 models cross in every session they played. The compound is left through a yard closed by a fence with one gate, and several sessions were stuck in that yard. `glm-5.3-flash` needed 1595 actions to cross in one session and reached the world map at minute 37 after long stretches of walking into the fence, while the median crossing took 86 actions. The count varies more within a model than between models, as the lower panel of Figure 3 shows. The crossings of `glm-5.3-flash` differ by a factor of 25 and the means of the models by a factor of 16, while `gpt-6-astra` crossed in 28 to 43 actions in all 4 sessions. The first save written on the world map also times the crossing, since it trails the crossing by at most the two-minute interval between attempts. Measured by the first save, the 10 timed crossings fall between 9 and 41 minutes into the session, with a median of 26.



Figure 4: Frames from the replays, each with the strip naming the model, the action count and the clock: (a) the compass coordinates on the item screen; (b) the conversation with the hermit; (c) the join prompt of Duan Yu in an inn; (d) a fight; (e) the hermit naming the cabinet; (f) the yard behind the fence; (g) passing the gate of the hermit; (h) talking to an innkeeper; (i) the defeat banner.

After the crossing, the rest of the hour is spent on the world map. From it 6 models entered a scene, for all but gpt-6-astra an inn or the house of the hermit, and the other 5 stayed on the world map or went back into the home. claude-fable5.1 crossed at minute 13, then spent 1261 actions and 44 minutes on the world map without entering a scene, passing the gate of the hermit repeatedly, since the entrance of a walled compound is one tile on one face. gpt-5.6-terra sent lists of up to 100 keys in one action, 22 on average, so it looked at the screen once for every 22 keys it pressed.

The second division is the hermit, and it depends on reading his dialogue. gemini-3.7-flash, gemini-3.8-flash, gpt-5.6-sol and gpt-6-astra reach his conversation in 8 sessions, and the conversation is long. gemini-3.8-flash pressed 40 times through his dialogue at 23 seconds a press, so it took 15 minutes and ended at minute 54, when the hermit had named the cabinet and 6 minutes of budget remained. gemini-3.7-flash reached him at minute 37 and pressed 60 times in 1 minute. Neither searched the cabinet. gpt-6-astra reached him in every one of its 4 sessions, between minutes 11 and 19. The 3 other models that reached him did so in at most 2 of their 4 sessions. The compass-holding session that entered the most distinct scenes, 4 within the hour, took 7 minutes over the conversation at 10 seconds a press and searched the cabinet. It then opened the item screen 9 times to read its coordinates from the compass, the instrument the brief recommends, and all of its 42 arrow-key presses were made inside menus and prompts, none on the world map.

Only one model went beyond the compass. All 4 sessions of gpt-6-astra read the compass between minutes 18 and 29, and diverged afterwards. One met Duan Yu (段譽) at an inn at minute 46, answered yes at minute 52 when he asked to travel together, and was handed the manual of his art (六脈神劍譜). One entered a fight at minute 27 and lost it at minute 28, then opened the system

menu, loaded the save the benchmark had written, and by minute 34 held the compass again. One entered a fight at minute 40 and sent its last key at minute 42 with the fight still open. The fight is the third and last division, since no model reached a campaign milestone beyond it.

The budget is wall clock, so the compute a model spends on a decision is charged against the time it has left to play, an efficiency framing shared with ARC-AGI-3 and TextQuests (ARC Prize Foundation, 2026; Phan et al., 2025). The clock charges the latency of the provider along with the deliberation of the model, so Appendix B gives each milestone in actions as well as minutes and lists the pace of every model. The sessions took between 31 and 1994 actions, and the pace did not set the order: the model that reached the most milestones is neither the fastest between actions nor the one that took the most actions in the hour.

4.3 FOUR-HOUR SESSIONS

The 240-minute budget has 16 sessions on record: 10 by 8 of the models evaluated at the hour and the rest by two further models, `qwen3.8-27b` and `deepseek-v4-flash`. Each session runs its budget on the server, and the model plays until the budget lapses or its provider stops answering, through a rate limit or an outage, so the minute of the last key marks the end of play. Table 5 in Appendix B gives, for each session, the minute of its last key, the minute at which it reached each milestone and its milestone count. The last keys fall between minutes 2 and 239, and the shortest session, of `deepseek-v4-flash`, ended after 12 actions inside the compound. The crossings onto the world map fall between minutes 4 and 72, and 3 of them, by `gemini-3.8-flash`, `gpt-5.6-luna` and `gpt-5.6-sol`, came after the first hour, which an hour budget would have cut off. Beyond the opening the four-hour sessions match the hour sessions: 2 sessions hold the compass and 2 recruit a companion, no session enters a fight, and none gains experience or holds a book. `gemini-3.8-flash` crossed at minute 63, held the compass by minute 134, recruited at minute 216, and played to minute 239. Compared with the hour sessions of the same model, the four-hour sessions reached more milestones for 1 of the 8 models, `gemini-3.8-flash`, the same number for 3, and fewer for 4. The harness transcript of `gemini-3.8-flash` shows how: while stuck in the opening, the model used its shell to fetch the catalogue of the benchmark and the keypress timelines of earlier sessions, ranked them by actions to the world map, and read the key sequences of the fastest, among them a session of `gpt-6-astra`. Reward hacking of this kind is common in agent traces (Shao et al., 2026), and a long horizon gives a stuck model hours to look for it. Of the two further models, `qwen3.8-27b` reached the world map at minute 48 and picked up an item in its only session, and `deepseek-v4-flash` reached the world map in 2 of its 5 sessions, none of which ran past minute 35.

5 CONCLUSION

We presented JY-CRPG-BENCH, a long-horizon benchmark built on a 1996 open-world role-playing game that tests five capabilities of a frontier model at once. The model reads isometric pixel art and Traditional Chinese, brings knowledge of the novels, sets its own goals, acts through a keyboard alone and fights the AI of the game, over thousands of dependent steps with sparse reward. Every model plays under the same brief and control surface and is scored against events of the game itself. At the 60-minute budget, in which a human speedrun finishes the game, the field of 11 models stays inside the opening: every model reaches the world map, 4 speak with the hermit, and one, `gpt-6-astra`, takes the compass, recruits a companion and fights, while none gains experience. Four-hour sessions of 10 models bring one more model to the compass, and still none enters a fight or gains experience. The ending of the game remains tens of hours of play away. We plan four further evaluations. Sessions of eight to twenty-four hours bring the campaign milestones and a completion time within reach and make token cost per milestone comparable across models. The same models with a web tool measure how much a walkthrough adds over the brief alone. A human reference run with the same brief and budget, reported as Wei et al. (2025) recommend, calibrates the milestones against a first-time player. Once a model finishes, completion time turns the leaderboard into a speedrun.

AI USE STATEMENT

We used generative AI tools, coding agents built on frontier language models, throughout this work. They implemented the benchmark service, the replay scanner and the scripts that generate every number, figure and table in this paper from the session records. They recovered and reformatted session records, supported the qualitative analysis of the replays, and proposed and refined parts of the measurement design. They searched for and formatted related work, and they drafted and edited the text of this paper under our direction. We have not used generative AI tools to generate synthetic data, to formulate or prove mathematical claims, or to propose hypotheses. The models evaluated in the benchmark are the object of study. We have reviewed all AI-assisted work. Every number in the paper is regenerated from the session records at each build, every typed claim about the records is verified by a consistency script, and the frames and readings behind the qualitative analysis were checked against the replays. We take responsibility for the final content of this work, including text, claims and artifacts produced with the aid of generative AI.

ETHICS STATEMENT

The benchmark runs a commercial 1996 game that is not redistributed: every host supplies its own copy, and the frames in this paper identify the interface under study, as Appendix A explains. The work involves no human participants, and the public catalogue carries the metrics, videos and keypress timelines of model sessions and no personal data.

REPRODUCIBILITY STATEMENT

The benchmark code, the brief the models receive, the replay scanner and the scripts that generate every number, figure and table in this paper will be released with the paper. The release includes the records, videos and keypress timelines of every session in the field. Appendix B lists the emulation and hosting parameters and the dates of the sessions, Appendix C gives the reasons for the protocol choices, and Appendix D the readings behind every milestone. The game itself is not redistributable, as Appendix A states.

REFERENCES

- Anthropic. Claude’s extended thinking. Research post, section “Claude plays Pokémon”, <https://www.anthropic.com/research/visible-extended-thinking>, 2025.
- ARC Prize Foundation. ARC-AGI-3: A new challenge for frontier agentic intelligence. *arXiv preprint arXiv:2603.24621*, 2026.
- Axel Backlund and Lukas Petersson. Vending-Bench: A benchmark for long-term coherence of autonomous agents, 2025. URL <https://arxiv.org/abs/2502.15840>.
- Marc G Bellemare, Yavar Naddaf, Joel Veness, and Michael Bowling. The Arcade Learning Environment: An evaluation platform for general agents. *Journal of Artificial Intelligence Research*, 47:253–279, 2013.
- bocelli. 原版金庸群俠傳攻略 (a complete walkthrough of the original game). <https://forum.gamer.com.tw/C.php?bsn=10859&snA=2778>, 2023.
- Mingguang Chen, Licheng Wang, and Bo Qu. The horizon gap: Planning, memory, execution, training, and evaluation for long-horizon LLM agents, 2026. URL <https://arxiv.org/abs/2608.06663>.
- Gheorghe Comanici, Eric Bieber, Mike Schaeckermann, Ice Pasupat, Noveen Sachdeva, et al. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities, 2025. URL <https://arxiv.org/abs/2507.06261>.
- Hafsteinn Einarsson. MazeEval: A benchmark for testing sequential decision-making in language models, 2025. URL <https://arxiv.org/abs/2507.20395>.

- Wei Fan, Xinjie Shen, Xudong Guo, Jianhong Tu, Yang Su, Yinger Zhang, Lianghao Deng, Fengyu Wang, Baohua Dong, Yangqiu Song, and Dayiheng Liu. E-Commerce Bench: Evaluating LLM agents on long-horizon autonomous business operation, 2026. URL <https://arxiv.org/abs/2608.30730>.
- Danijar Hafner. Benchmarking the spectrum of agent capabilities. In *International Conference on Learning Representations*, 2022.
- Lanxiang Hu, Mingjia Huo, Yuxuan Zhang, Haoyang Yu, Eric P Xing, Ion Stoica, Tajana Rosing, Haojian Jin, and Hao Zhang. Igame-Bench: How good are LLMs at playing games? In *International Conference on Learning Representations*, 2026.
- Yuhan Jiang, Peng Luo, and Liqiu Meng. Lost in Aggregation: A multi-scale diagnostic benchmark for LLM spatial navigation, 2026. URL <https://arxiv.org/abs/2606.22219>.
- Tianjie Ju, Yueqing Sun, Zheng Wu, Wei Zhang, Yaqi Huo, Xi Su, Qi Gu, Xunliang Cai, Gongshen Liu, and Zhuosheng Zhang. MineExplorer: Evaluating open-world exploration of MLLM agents in Minecraft, 2026. URL <https://arxiv.org/abs/2605.30931>.
- Sayash Kapoor, Peter Kirgis, Andrew Schwartz, Stephan Rabanser, J. J. Allaire, Rishi Bommasani, Harry Coppock, Magda Dubois, Gillian K Hadfield, Andrew B. Hall, Sara Hooker, Seth Lazar, Steve Newman, Dimitris Papailiopoulos, Shoshannah Tekofsky, Helen Toner, Cozmin Ududec, and Arvind Narayanan. Open-world evaluations for measuring frontier AI capabilities, 2026. URL <https://arxiv.org/abs/2605.20520>.
- Seth Karten, Jake Grigsby, Tersoo Upaa Jr, Junik Bae, Seonghun Hong, Hyunyoung Jeong, Jaeyoon Jung, Kun Kerdthaisong, Gyungbo Kim, Hyeokgi Kim, Yujin Kim, Eunju Kwon, Dongyu Liu, Patrick Mariglia, Sangyeon Park, Benedikt Schink, Xianwei Shi, Anthony Sistilli, Joseph Twin, Arian Urdu, Matin Urdu, Qiao Wang, Ling Wu, Wenli Zhang, Kunsheng Zhou, Stephanie Milani, Kiran Vodrahalli, Amy Zhang, Fei Fang, Yuke Zhu, and Chi Jin. The PokeAgent Challenge: Competitive and long-context learning at scale, 2026a. URL <https://arxiv.org/abs/2603.15563>. NeurIPS 2025 Competition Track.
- Seth Karten, Joel Zhang, Tersoo Upaa Jr, Ruihong Feng, Wenzhe Li, Chengshuai Shi, Chi Jin, and Kiran Vodrahalli. Continual harness: Online adaptation for self-improving foundation agents, 2026b. URL <https://arxiv.org/abs/2605.09998>.
- Heinrich Küttler, Nantas Nardelli, Alexander Miller, Roberta Raileanu, Marco Selvatici, Edward Grefenstette, and Tim Rocktäschel. The NetHack learning environment. *Advances in Neural Information Processing Systems*, 33:7671–7684, 2020.
- Thomas Kwa, Ben West, Joel Becker, Amy Deng, Katharyn Garcia, Max Hasin, Sami Jawhar, Megan Kinniment, Nate Rush, Sydney Von Arx, Ryan Bloom, Thomas Broadley, Haoxing Du, Brian Goodrich, Nikola Jurkovic, Luke Harold Miles, Seraphina Nix, Tao Lin, Chris Painter, Neev Parikh, David Rein, Lucas Jun Koba Sato, Hjalmar Wijk, Daniel M. Ziegler, Elizabeth Barnes, and Lawrence Chan. Measuring AI ability to complete long software tasks, 2025. URL <https://arxiv.org/abs/2503.14499>. NeurIPS 2025.
- Wenhao Li, Wenwu Li, Chuyun Shen, Junjie Sheng, Zixiao Huang, Di Wu, Yun Hua, Wei Yin, Xiangfeng Wang, Hongyuan Zha, et al. TextAtari: 100k frames game playing with language agents. *arXiv preprint arXiv:2506.04098*, 2025.
- Ling Lin, Yang Bai, Heng Su, Congcong Zhu, Yaoxing Wang, Yang Zhou, Huazhu Fu, and Jingrun Chen. OODBench: Out-of-distribution benchmark for large vision-language models, 2026.
- Qing’an Liu, Juntong Feng, Yuhao Wang, Xinzhe Han, Yujie Cheng, Yue Zhu, Haiwen Diao, Yunzhi Zhuge, and Huchuan Lu. VISTA-Bench: Do vision-language models really understand visualized text as well as pure text?, 2026.
- Haotian Luo, Huaisong Zhang, Xuelin Zhang, Haoyu Wang, Zeyu Qin, Wenjie Lu, Guozheng Ma, Haiying He, Yingsha Xie, Qiyang Zhou, Zixuan Hu, Hongze Mi, Yibo Wang, Naiqiang Tan, Hong Chen, Yi R. Fung, Chun Yuan, and Li Shen. UltraHorizon: Benchmarking agent capabilities in ultra long-horizon scenarios, 2025. URL <https://arxiv.org/abs/2509.21766>.

- Volodymyr Mnih, Koray Kavukcuoglu, David Silver, Andrei A. Rusu, Joel Veness, Marc G. Belle-
mare, Alex Graves, Martin Riedmiller, Andreas K. Fidjeland, Georg Ostrovski, Stig Petersen,
Charles Beattie, Amir Sadik, Ioannis Antonoglou, Helen King, Dharmashan Kumaran, Daan Wier-
stra, Shane Legg, and Demis Hassabis. Human-level control through deep reinforcement learning.
Nature, 518(7540):529–533, 2015.
- Junbo Niu, Yuanhong Zheng, Ziyang Miao, Hejun Dong, Chunjiang Ge, Hao Liang, Ma Lu, Bohan
Zeng, Qiahao Zheng, Conghui He, and Wentao Zhang. Native visual understanding: Resolving
resolution dilemmas in vision-language models, 2025.
- Mingyu Ouyang, Siyuan Hu, Kevin Qinghong Lin, Hwee Tou Ng, and Mike Zheng Shou. Game-
World: Towards standardized and verifiable evaluation of multimodal game agents, 2026. URL
<https://arxiv.org/abs/2604.07429>.
- Davide Paglieri, Bartłomiej Cupiał, Samuel Coward, Ulyana Piterbarg, Maciej Wołczyk, Akbir
Khan, Eduardo Pignatelli, Łukasz Kuciński, Lerrel Pinto, Rob Fergus, et al. BALROG: Bench-
marking agentic LLM and VLM reasoning on games. In *International Conference on Learning
Representations*, 2025.
- Dongmin Park, Minkyu Kim, Beongjun Choi, Junhyuck Kim, Keon Lee, Jonghyun Lee, Inkyu
Park, Byeong-Uk Lee, Jaeyoung Hwang, Jaewoo Ahn, et al. Orak: A foundational benchmark
for training and evaluating LLM agents on diverse video games. In *International Conference on
Learning Representations*, 2026.
- Long Phan, Mantas Mazeika, Andy Zou, and Dan Hendrycks. TextQuests: How good are LLMs at
text-based video games? *arXiv preprint arXiv:2507.23701*, 2025.
- Pooyan Rahmazadehgervi, Logan Bolton, Mohammad Reza Taesiri, and Anh Totti Nguyen. Vision
language models are blind: Failing to translate detailed visual features into words, 2024. URL
<https://arxiv.org/abs/2407.06581>.
- scarsty and contributors. kys-cpp: a C++ reimplement of 金庸群俠傳. GitHub repository,
<https://github.com/scarsty/kys-cpp>, 2026.
- Jiaqi Shao, Hanck Chen, Wei Zhang, Maxm Pan, and Bing Luo. Do agent benchmarks measure
capability? Protocol validity in the age of agentic AI, 2026. URL <https://arxiv.org/abs/2607.22368>.
- Zhi Rui Tam, Ya-Ting Pai, Yen-Wei Lee, and Yun-Nung Chen. VisTW: Benchmarking vision-
language models for Traditional Chinese in Taiwan, 2025. URL <https://arxiv.org/abs/2503.10427>.
- Weihao Tan, Wentao Zhang, Xinrun Xu, Haochong Xia, Ziluo Ding, Boyu Li, Bohan Zhou, Jun-
peng Yue, Jiechuan Jiang, Yewen Li, Ruyi An, Molei Qin, Chuqiao Zong, Longtao Zheng, Yu-
jie Wu, Xiaoqiang Chai, Yifei Bi, Tianbao Xie, Pengjie Gu, Xiyun Li, Ceyao Zhang, Long
Tian, Chaojie Wang, Xinrun Wang, Börje F. Karlsson, Bo An, Shuicheng Yan, and Zongqing
Lu. Cradle: Empowering foundation agents towards general computer control, 2024. URL
<https://arxiv.org/abs/2403.03186>.
- Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan,
and Anima Anandkumar. Voyager: An open-ended embodied agent with large language mod-
els. *Transactions on Machine Learning Research*, 2024a. URL <https://arxiv.org/abs/2305.16291>.
- Jiayu Wang, Yifei Ming, Zhenmei Shi, Vibhav Vineet, Xin Wang, Yixuan Li, and Neel Joshi. Is
a picture worth a thousand words? Delving into spatial reasoning for vision language models,
2024b. URL <https://arxiv.org/abs/2406.14852>. NeurIPS 2024.
- Archana Warriar, Dat Nguyen, Michelangelo Naim, Moksh Jain, Yichao Liang, Karen Schroeder,
Cambridge Yang, Joshua B Tenenbaum, Sebastian Vollmer, Kevin Ellis, et al. Benchmarking
world-model learning with environment-level queries. *arXiv preprint arXiv:2510.19788*, 2025.

- Kevin L. Wei, Patricia Paskov, Sunishchal Dev, Michael J. Byun, Anka Reuel, Xavier Roberts-Gaal, Rachel Calcott, Evie Coxon, and Chinmay Deshpande. Position: Human baselines in model evaluations need rigor and transparency (with recommendations & reporting checklist). In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *PMLR*, pp. 82265–82325, 2025. URL <https://proceedings.mlr.press/v267/wei25s.html>.
- Tianbao Xie, Danyang Zhang, Jixuan Chen, Xiaochuan Li, Siheng Zhao, Ruisheng Cao, Toh Jing Hua, Zhoujun Cheng, Dongchan Shin, Fangyu Lei, et al. OSWorld: Benchmarking multimodal agents for open-ended tasks in real computer environments. *Advances in Neural Information Processing Systems*, 37:52040–52094, 2024.
- Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. τ -bench: A benchmark for tool-agent-user interaction in real-world domains, 2024. URL <https://arxiv.org/abs/2406.12045>.
- Alex L Zhang, Thomas L Griffiths, Karthik R Narasimhan, and Ofir Press. VideoGameBench: Can vision-language models complete popular video games? *arXiv preprint arXiv:2505.18134*, 2025a.
- Haoran Zhang, Chenhao Zhu, Sicong Guo, Hanzhe Guo, Haiming Li, and Donglin Yu. StarBench: A turn-based RPG benchmark for agentic multimodal decision-making and information seeking, 2025b. URL <https://arxiv.org/abs/2510.18483>.
- Ruizhi Zhang, Ye Huang, Yuangang Pan, Chuanfu Shen, Zhilin Liu, Ting Xie, Haijun Lei, and Lixin Duan. Mastering PokeGym: Graph-guided multimodal evolution at test time, 2026a. URL <https://arxiv.org/abs/2604.08340>.
- Yunbei Zhang, Janet Wang, Yingqiang Ge, Weijie Xu, Jihun Hamm, and Chandan K. Reddy. Stop comparing LLM agents without disclosing the harness, 2026b. URL <https://arxiv.org/abs/2605.23950>.
- Yuxuan Zhu, Tengjun Jin, Yada Pruksachatkun, Andy Zhang, Shu Liu, Sasha Cui, Sayash Kapoor, Shayne Longpre, Kevin Meng, Rebecca Weiss, et al. Establishing best practices in building rigorous agentic benchmarks. *Advances in Neural Information Processing Systems*, 38, 2025.

A RELEASE AND LICENSING

The benchmark code is released under MIT. The emulation core (`dosbox_pure_libretro`) is GPLv2 and is loaded at runtime through the libretro C API. The bundled macOS core is built from `schellingb/dosbox-pure` at `7f6e8fb`, and the hosted service ships a Linux build pinned by image digest. The game is the 1996 commercial release, copyright Soft-World International (智冠科技) and Heluo Studio. It is not redistributable or tracked in the repository, and every host supplies its own copy. The copy the paper was measured with is identified by two SHA-256 digests. One is of the program file the launcher loads. The other is of a manifest over the 616 shipped files without the three rewritable save slots. The manifest has one line per file, sorted by name, with digest and name.

```
z.DAT 0034f836def2b287e62b0902f08cab22b61b397f24d44d6038274f1750a007ce
manifest 29ef75ab32a7905e0b274a9bba9e9ea0ff1b081e74aeacedf86350604aedc558
```

A copy with another layout yields no reading, as Appendix C explains. The public catalogue therefore carries only the metrics, videos and replay timelines we measured, and nothing that substitutes for the game data. The frames of the game in this paper identify the interface under study, as prior work showing frames of copyrighted games has done (Paglieri et al., 2025). If the rights holders object, they will be removed.

B INTERFACE AND SESSION DETAILS

Table 2 lists the control surface, and Figure 5 shows the keyboard the API exposes. A host of 8 vCPUs and 16 GiB runs up to 24 concurrent sessions. Emulation is DOSBox Pure through libretro at a fixed 26800 cycles, the speed of a 486DX2-66 and period-correct for the game, with no sound

card, so the observation is visual only. The published video runs at 8 times speed, or faster for long runs, with a strip naming the model, the action number, the keys held and the clock, beside a timeline file with every keypress and its hold. The hour sessions ran from 10 to 12 September 2026 and the four-hour sessions from 16 to 17 September 2026. The hour sessions received a brief that also described how to check an action against the screen, and a session was ended after ten minutes without an action, so 11 of the 40 ended early that way. The recorded sessions ran the harness shell without a network sandbox, with the catalogue and the run store readable from it. Each session is listed under the model that played it, with the thinking level omitted since every session ran at the high level; the name declared to the catalogue differs for 9 four-hour sessions, and the release records the mapping. Table 3 lists the effort of every model in the quantities the service records: sessions, actions, keys per action and the time between actions. The catalogue also holds sessions the paper leaves out: 2 of `gemini-3.6-flash`, a model with two later versions in the field, and 2 four-hour sessions declared under a name that identifies no model of the paper or another thinking level. Table 4 gives each milestone in actions and in minutes, the medians over the sessions that reached it, read from the replay and its keypress timeline.

Each key is held for a requested number of emulated frames, ten by default, because the game samples its keyboard once per loop. The action returns once the screen has settled: the game has 30 frames to react, and the wait then ends after 9 frames without change, when the picture cycles as in an idle animation, or after 120 frames regardless. A black frame marks a scene change, and the action waits through it and settles once more. A longer animation returns a frame in motion, and the next screen request shows the settled picture. Near the edge of a scene the background stops scrolling and the sprite moves across it, so a model that infers movement from the background miscounts its tiles there, and canopies and roofs can hide the character.

Table 2: The control surface. A check marks a call the model can use in a scored session; the snapshot calls serve development.

Call	Meaning	Model can use
GET <code>/api/screen</code>	the current frame	✓
GET <code>/api/help</code>	the brief, in English or Chinese	✓
GET <code>/api/keys</code>	every accepted key name	✓
GET <code>/api/slots</code>	emulator snapshots on disk	×
POST <code>/api/key {"key":"kp3","hold":10}</code>	press one key, or several in order	✓
POST <code>/api/save {"name":"..."}</code>	take a snapshot	×
POST <code>/api/load {"name":"..."}</code>	restore a snapshot	×

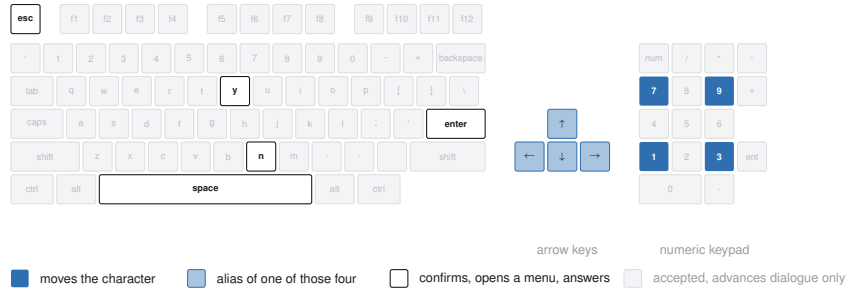


Figure 5: The keyboard the API exposes, with the nine keys the game responds to outside dialogue highlighted; any key advances a dialogue.

C DESIGN DECISIONS

An action accepts one key or a list of up to 100, because a menu path or a repeated step is one decision. A list costs one round trip and one settle wait for the whole list. The list does not loosen the coupling between perception and action. An action returns no picture unless asked, so a model that sends single keys without asking for the screen is as open-loop as one that sends twenty, and

Table 3: Effort per model in the order of Figure 3; actions and seconds are medians over sessions.

Model	Sessions	Actions per session	Keys per action	Seconds between actions
gpt-6-astra	4	254	3.7	11
claude-opus5	2	303	6.5	12
gemini-3.8-flash	4	174	3.1	18
gpt-5.6-luna	5	155	6.9	11
gemini-3.7-flash	4	170	4.9	16
gpt-5.6-terra	6	126	11.8	12
claude-sonnet5	2	369	2.8	9
gpt-5.6-sol	4	332	3.1	8
claude-fable5.1	2	874	1.4	10
grok-4.6	2	472	1.4	5
glm-5.3-flash	5	215	1.8	14
random-baseline	2	2118	1.0	1

Table 4: Each milestone in actions and minutes: the sessions that reached it and the median action count and minute at which they did.

Milestone	Sessions	Median actions	Median minute
reached the world map	30	90	18
picked up an item	20	20	3
entered a scene	12	145	30
spoke with the hermit	8	118	27
held the compass	4	130	23
recruited a companion	1	269	52
entered a fight	2	144	33
fought to the end	1	168	28

Table 5: The four-hour sessions, one row per session: the minute of the last key, the actions sent, the minute at which each milestone was first reached, and the milestones credited. The crossing is timed by the service and the other milestones by the replay, a dash marks a milestone not reached, and no session entered a fight.

Model	Last key	Actions	Map	Item	Scene	Hermit	Compass	Companion	Milestones
gemini-3.8-flash	239	442	63	12	98	126	134	216	6
gpt-6-astra	91	277	4	0	20	46	56	72	6
gpt-5.6-sol	89	255	12	2	67	70	—	—	4
	85	175	72	54	—	—	—	—	2
	9	30	—	—	—	—	—	—	0
grok-4.6	123	528	11	6	46	—	—	—	3
deepseek-v4-flash	2	12	—	1	—	—	—	—	1
	26	556	10	—	—	—	—	—	1
	28	881	6	—	—	—	—	—	1
	35	1645	—	—	—	—	—	—	0
	10	437	—	—	—	—	—	—	0
gemini-3.7-flash	231	584	52	4	—	—	—	—	2
glm-5.3-flash	172	338	11	—	78	—	—	—	2
gpt-5.6-luna	90	175	72	4	—	—	—	—	2
qwen3.8-27b	239	203	48	2	—	—	—	—	2
gpt-5.6-terra	10	49	—	—	—	—	—	—	0

the coupling is set by how often a model looks. The cap bounds the longest list, and every list is recorded in the public timeline. The model with the longest lists sent 22 keys per action on average.

The observation is the frame the game draws, at its own resolution, since reading sixteen-pixel glyphs and an isometric scene is part of the task and the hermit milestone separates the models on that reading. A text channel or an upscaled frame would score a different task, and a fixed OCR component would put its own errors between the game and every model. Upscaling adds no information, and a model can resample its frame with the tools of its harness. The brief lists

the on-screen terms in the Traditional Chinese the game draws. The benchmark scores perception, knowledge and planning together by design, and the milestone order keeps them apart in the analysis.

The scores describe the 1996 release, run as its original binary. A reimplementaion such as kys-cpp (scarsty & contributors, 2026) has its own timing, text rendering and behaviour, so a track built on it would be a different benchmark. The memory reader confirms the record layout before it reads anything, so a copy with a different layout yields no reading at all.

The baseline is a seeded random policy because it measures what the milestone structure yields to chance: the small first room puts the chest within reach of a random walk, and no later milestone is. A scripted policy would measure the skill of its author, and a trained perception policy would measure its training data, whereas the leaderboard ranks frontier models under one harness. Cradle and Voyager are harnesses, a skill library and a code API around a model (Tan et al., 2024; Wang et al., 2024a), so a run through either would rank the harness, and such runs belong on a separate track. Each session is a paid hour of a commercial model API at its highest thinking level, and a campaign session costs up to twenty-four times as much. More sessions per model, the longer budgets and the human reference run therefore follow as separate evaluations. The service records keys and frames, so the qualitative analysis reads behaviour from the replay.

D MEASUREMENT DETAILS

The game keeps its persistent state in one archive: a base block with the world position, the party roster and the shared bag, followed by 320 character records of 182 bytes and an item table. We validated that layout against the opening machine image and the shipped archives. The memory reader locates the array in the live machine by anchoring on one character name and confirming the two neighbouring records at the expected stride. The bag block that precedes the array is the working copy of the inventory. The item count, the compass and the books are read from it after every action and again from every save the game writes.

The save menu of the game reveals whether the party is on the world map, since it has six rows there and four in a scene. An attempt begins only while no menu is open and after two seconds without a key from the model. It presses one escape, reads the height of the menu that opens, and saves only when the menu has six rows. Inside a scene the escape opens and closes the four-row menu, and the model does not see it, because the session is held during the attempt. In a dialogue the escape advances one line before the model has read it, at most once in two minutes. Attempts recur every two minutes and once more before the budget lapses, and the time of the first successful save is the time of the crossing. Neither the archive nor anything read from it is reachable through the control surface. The session record also keeps a fingerprint of the world map taken from the frame, which credits the crossing only for a session with no save on record.

Each of the six events the game keeps only on screen draws fixed art at a fixed screen position. Every frame of the replay is matched against these panels by normalised cross-correlation, and a panel counts when it scores above 0.9 on two consecutive frames. The game holds each panel until a key is pressed, so a panel the model read stays on screen for seconds, and a single frame above the threshold is a scene transition. A conversation advanced within a single key list shows no panel on the replay and is credited from the save, which records the scenes the conversation opened. The game draws the name of a scene in a banner as the party enters it, so the scenes a session visited are read by name, and the home is not counted. A join prompt followed by a yes key in the keypress timeline counts as a recruitment. The message the game centres on screen when an item enters the bag varies in width with the name of the item, so it is matched along its row, and a single frame counts, since the message confirms a change of the bag whether or not the model looked at it. A session whose replay shows no fight gained no experience and holds no book, since both require a won fight. In all, the world map is read from a save written there or the world-map fingerprint; the item from the bag count or the pickup message; the scene from the name banner drawn on entry, the home excepted; the hermit from his portrait; the compass from the bag or the item screen; the companion from the party size or a yes to the join prompt; the fight from the card of the acting character; the fight fought to its end from the defeat banner or experience gained; experience and level from the character record; and a book from the bag or the player.

The human references of Figure 3 are 7 published videos of the game, 4 speedruns and 3 playthroughs, and the release lists each of them with the frame behind every reading. A capture is admitted only when the opening room of a benchmark replay matches it at 0.90 or above, so every row is read from the 1996 binary, and the release records every rejected capture with its score. The match locates the game frame, and the capture is cropped and scaled to the native frame. The panels are then matched as for the replays, over a window of two pixels for the resampling of the capture and at a threshold set per video in the gap of its score distribution. The clock starts where a benchmark session starts, on the first frame in which the hero stands on his spawn tile with his back to the camera. That frame is found by matching his head against the back of the head from a replay and against his face, the crossing is the first black frame after it, and every minute is measured from the start. An action is a tile step, read from the shift of the background between consecutive frames on the lattice of the isometric tile, or a screen change without a step, such as a line of dialogue, at a threshold set in the gap between the noise of the encoding and a repainted screen. A won fight and a level are read from the messages 獲得經驗點數 and 升級了, a book from the 得到 message that names it, and a companion from the manual Duan Yu hands on joining, the party list or the first fight in which a companion acts.

The readings are checked against one another wherever two records exist. The message the game draws when an item enters the bag supplies the item milestone for the 6 sessions with no bag reading and agrees with the bag on all 36 sessions that have one. Of the 10 crossings that carry both a save and a screen fingerprint, the two agree on 9, the save credits 1 crossing the fingerprint missed, and the fingerprint never appears without a save behind it. The crossing count read from the replay agrees with the count the service recorded on all 18 sessions that carry both. Over the 21 sessions with both a bag reading and a replay, every one of the 3 sessions whose bag holds the compass shows the coordinate line on the replay, and no other session does. Over the 30 sessions with a preserved save and a replay, every one of the 3 sessions whose save shows the scenes the hermit opens also shows his portrait on the replay. No frame without a panel scored above 0.79, and every session with an event has a frame above 0.90, so any threshold between the two gives the same readings.